

## NeoVim Cheat - Normal Mode

The diagram illustrates Emacs keyboard shortcuts, organized into three rows of colored boxes. Each box contains a key symbol, its function, and a line number (e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9).

**Row 1: Basic Editing**

- ~** toggle case
- !** 1
- @** play macro 2
- #** find (rev) identifier 2
- \$** eol 2
- %** goto match 2
- ^** "soft" bol 2
- &** repeat :s 2
- \*** find identifier 2
- (** begin sentence 2
- )** end sentence
- \_** "soft" bol down previous line
- +** next line
- =** auto format 9
- Backsp** delete left char

**Row 2: Text Manipulation**

- Q** play last macro
- W** next WORD
- E** end WORD
- R** replace mode
- T** till char backward
- Y** yank to eol
- U** undo line
- I** insert at bol
- O** open above
- P** paste prev line 1
- {** prev aerial section
- }** next aerial section
- |** bol / go to column
- q** record macro
- w** next word
- e** end word
- r** replace char
- t** till char forward
- y** yank 1,7 8,9
- u** undo
- i** insert mode
- o** open below
- p** paste next line 1
- [** previous 3
- ]** next 3
- \** resume telescope

**Row 3: Navigation and Other**

- Caps = Esc** normal mode
- A** append at eol
- S** delete line
- D** delete to eol
- F** find char backward
- G** eof / goto line
- H** screen top
- J** join lines
- K** help / doc
- L** screen bottom
- :** command line
- "** choose register 1
- a** append
- s** delete char
- d** delete 1,7 8,9
- f** find char forward
- g** extra cmds 5
- h** ◀
- j** ▼
- k** ▲
- l** ▶
- ;** hop char full screen
- '** goto mark bol
- Enter** change word

**Row 4: Navigation and Other (Continued)**

- Z** scroll cmds 4
- X** back-space
- C** delete to eol
- V** visual lines
- B** back WORD
- N** previous (find)
- M** screen middle
- <** un-indent 7,9
- >** indent 7,9
- ?** find (rev)
- Z** delete char
- X** delete char
- C** change 1,7 8,9
- V** visual mode 7
- b** back word
- n** next (find)
- m** set mark 6
- ,**
- .** repeat cmd
- /** find

**Legend:**

- bol** = beginning of line
- eol** = end of line,

bol = beginning of line  
eol = end of line,

|          |                                                                     |
|----------|---------------------------------------------------------------------|
| motion   | moves the cursor, or defines the range for an operator              |
| command  | direct action command, if <b>red</b> it enters insert mode          |
| operator | requires a motion afterwards, operates between cursor & destination |
| extra    | special functions, requires extra input                             |
| q.       | commands with a dot need a char argument afterwards                 |

### main command line commands ('ex'):

```

:w (save), :q (quit) :wq (save&quit), :q! (quit w/o saving)
:e f (open file f), :vsp f (open file in vertical split)
:%s/x/y/gc (replace 'x' by 'y' filewide, but ask),
:%s/\s+$/ / (remove all trailing spaces),
:h (help in neovim), :new (new file in neovim)
:[number] (go to line [number])
:reg (show content of registers)

```

**visual mode:**

- operator use selection instead of motion
- o: exchange cursor position with start of selection
- J: join all selected lines
- S: add surrounding delimiter

**notes:**

- (1) use "x before a yank/paste/del command to use that register ('clipboard') (x=a..z,\*) (e.g.: "ay\$ to copy rest of line to register 'a')
- (2) type in a number before any action to repeat it that number of times (e.g.: 2p, d2w, 5i, d4j)
- (3) [m / ]m - go to previous / next method start [M / ]M - go to previous / next method end [t / ]t - go to previous / next TODO or NOTE [e / ]e - go to previous / next diagnostic [g / ]g - go to previous / next git hunk
- (4) zz: scroll cursor to center, zt: top, zb:bottom
- (5) gg: top of file, gi: go to last insertion and switch to insert mode gv: reselect last visual selection gm: middle of screen line, gM: middle of line ge: previous end of word gn / gN: go to next/previous find and select gt / gT: go to next/previous tab
- (6) m, - set next available mark (alphabetical) m; - toggle mark at current line m] and m[ - go to next/previous mark dmx and dm<space> - delete mark x / delete all marks
- (7) after v, y, c, d, <, > use a (around) or i (inside) followed by ( ), [ ], { }, < , > , ' ' , ` ` , w (word), p (paragraph) f (function), c (class), s (statement), l (loop) b (block), m (method call), a (argument), k (comment) ys[motion]{new} - add surrounding {new} to {motion} cs[old]{new} - change surrounding {old} with {new} ds[old] - delete surrounding {old} yss[new] - add surrounding {new} to line [space]b - change brackets (same as csb) {new/old} = ( ), [ ], { }, < , > , ' ' , ` ` , s (any of those) b (brackets: ), ], > , q (quotes: " , ' , ` ` , t (html-tags), f (function), i (individual start and end)
- (9) duplicate operator to act on current line (dd: delete line, cc: change line, yy: yank line, >>: indent line)

# NeoVim Cheat Sheet - [Space]-Commands

|                        |        |        |        |         |        |        |        |        |                       |                       |     |     |        |
|------------------------|--------|--------|--------|---------|--------|--------|--------|--------|-----------------------|-----------------------|-----|-----|--------|
| ~<br>change inside ... | !<br>1 | @<br>2 | #<br>3 | \$<br>4 | %<br>5 | ^<br>6 | &<br>7 | *<br>8 | ( change inside (...) | ) change inside (...) | _ - | + = | Backsp |
|------------------------|--------|--------|--------|---------|--------|--------|--------|--------|-----------------------|-----------------------|-----|-----|--------|

|                             |                  |                                           |                            |                             |        |                       |                             |        |               |                                                |                                                |                                 |
|-----------------------------|------------------|-------------------------------------------|----------------------------|-----------------------------|--------|-----------------------|-----------------------------|--------|---------------|------------------------------------------------|------------------------------------------------|---------------------------------|
| Q<br>q write buf & quit win | W<br>W write buf | E render cur error<br>e render next error | R<br>r• rename / rust cmds | T<br>t peek type definition | Y<br>y | U<br>u list uses/refs | I<br>i list def, type, refs | O<br>o | P<br>p format | { change inside (...)<br>[ change inside (...) | } change inside (...)<br>] change inside (...) | nvim mappings<br>\ nvim options |
|-----------------------------|------------------|-------------------------------------------|----------------------------|-----------------------------|--------|-----------------------|-----------------------------|--------|---------------|------------------------------------------------|------------------------------------------------|---------------------------------|

|            |                             |                                       |                        |                   |                  |        |        |        |                         |        |                                                |                              |
|------------|-----------------------------|---------------------------------------|------------------------|-------------------|------------------|--------|--------|--------|-------------------------|--------|------------------------------------------------|------------------------------|
| Caps = Esc | A• chatgpt<br>a code action | S• spectre global<br>S• spectre local | D<br>d peek definition | F<br>f• telescope | G<br>g• git cmds | H<br>h | J<br>j | K<br>k | L<br>l toggle lsp lines | :<br>; | " change inside "..."<br>' change inside '...' | Enter<br>change inside (...) |
|------------|-----------------------------|---------------------------------------|------------------------|-------------------|------------------|--------|--------|--------|-------------------------|--------|------------------------------------------------|------------------------------|

|        |        |                                      |        |                   |                 |                                                |                          |                          |                                      |
|--------|--------|--------------------------------------|--------|-------------------|-----------------|------------------------------------------------|--------------------------|--------------------------|--------------------------------------|
| Z<br>z | X<br>x | C change quotes<br>c change brackets | V<br>v | B<br>b• debugging | N<br>n• neotest | M select all visual-mul<br>m search visual-mul | < change inside <,><br>, | > change inside <,><br>. | ? keymap overview<br>/. cheat sheets |
|--------|--------|--------------------------------------|--------|-------------------|-----------------|------------------------------------------------|--------------------------|--------------------------|--------------------------------------|

## cheat sheets:

[Space]/n - normal\_mode.pdf  
[Space]/s - space.pdf  
[Space]/c - ctrl.pdf  
[Space]/a - alt.pdf

**lsp**

commands that uses the language server protocol

**edit**

direct action command, if **red** it enters insert mode

**open**

open a sidebar or floating window

**extra**

cargo / rust commands  
git commands

**f•**

commands with a dot need need additional inputs

## spectre:

[Space]SR - search & replace in project (normal/visual)  
[Space]SW - search current word in project (normal)  
[Space]sr - search and replace in file (normal/visual)  
[Space]sr - search current word in file (normal mode)

## chatgpt:

[Space]Ac - open chat window  
[Space]Ae - edit with instruction  
[Space]Ag - grammar correction  
[Space]At - translate  
[Space]Ak - keywords  
[Space]Ad - create docstring  
[Space]Aa - add tests  
[Space]Ao - optimize code  
[Space]As - summarize  
[Space]Af - fix bug  
[Space]Ax - explain code  
[Space]Al - readability analysis

## git:

[Space]gb - toggle git blame  
[Space]gs - toggle git signs (line diff in signcolumn)  
[Space]gw - toggle git inline changes (word diff)  
[Space]gl - open git log of current file to open vimdiff  
[Space]gh / [Space]gH - open git DiffviewFileHistory current file / all  
[Space]gd - open git Diffview against current index (close with Ctrl-c)

## rename and rust / cargo commands:

[Space]rn - rename symbol (refactoring)  
[Space]rt - run test under cursor  
[Space]rT - run all tests  
[Space]rb - cargo build  
[Space]rc - cargo clippy  
[Space]rr - cargo run  
[Space]rd - open docs in browser  
[Space]re / [Space]cE - explain next / current error

## visual-multi (multiple cursors):

Ctrl-Up/Down/Left/Right - toggle cursor at current location and move  
[Tab] - switch between cursor and extend mode  
Up/Down/Left/Right - move independent of other cursors  
Shift-Up/Down/Left/Right - switch to extend mode and select in direction  
n/N - go to next/previous match and add a cursor there  
q/Q - remove current cursor and go the next/previous match  
]/[ - go the next/previous cursor  
o - switch to other side of selection (in extend mode)  
[Space][Space]a - algin at cursors  
[Space][Space]C - case conversion menu  
[Space][Space]n / [Space][Space]0n - add increasing numbers starting at 1/0

## telescope:

[Space]ff - find files  
[Space]FF - find files (incl. hidden/ignored)  
[Space]fo - select old / recent files  
[Space]fb - select buffers  
[Space]fg - live grep in working directory  
[Space]fu - fuzzy find in current file  
[Space]fp - select from yanks for pasting  
[Space]fj - select jumplist  
[Space]fm - select marks  
[Space]fr - select from register  
[Space]f/ - select from search history  
[Space]fc - select from command history  
[Space]f: - find commands  
[Space]fh - find help tags  
[Space]fk - find keymaps  
[Space]fe - list diagnostics  
[Space]fv - list symbols (variable names) of current file  
[Space]fV - list symbols (variable names) of workspace  
[space]fa - open aerial picker  
[Space]fq - select quickfix  
[Space]fs - select spell suggestions  
[Space]ft - select TODOs and NOTES  
[Space]fzc - select colorscheme  
[Space]fzo - find vim-options  
[Space]fzh - find highlights  
[Space]fza - find autocommands  
[Space]fz - find telescope picker

Ctrl-U / Ctrl-D - scroll preview window up/down (normal mode in telescope)  
Ctrl-V / Ctrl-T (or v / t in normal mode) - open in new vsplit / tab

## debugging:

[Space]bp - toggle breakpoint  
[Space]bs - start new debugging session  
[Space]bt - terminate  
[Space]bc - continue to next breakpoint  
[Space]bn - next step / step over  
[Space]bi - step in  
[Space]bo - step out  
[Space]bu - toggle dap-ui

## neotest:

[Space]bt - run test under cursor  
[Space]nf - run all tests in file  
[Space]na - run all tests in project  
[Space]ns - stop running tests  
[Space]np - toggle output panel  
[Space]nn - toggle summary panel  
[Space]nw - toggle watch under cursor  
[Space]nc - clear output panel

# NeoVim Cheat Sheet - [Ctrl]-Commands

|                       |                              |                                             |                                       |                               |                     |                                       |              |                                            |                                        |       |   |                              |    |
|-----------------------|------------------------------|---------------------------------------------|---------------------------------------|-------------------------------|---------------------|---------------------------------------|--------------|--------------------------------------------|----------------------------------------|-------|---|------------------------------|----|
| `                     | 1                            | 2                                           | 3                                     | 4                             | 5                   | 6                                     | 7            | 8                                          | 9                                      | 0     | - | =                            | Bs |
| Q                     | W                            | E                                           | R                                     | T                             | Y                   | U                                     | I            | O                                          | P                                      | [     | ] | \                            |    |
| insert<br>literal key | del next<br>word             | scroll line<br>up<br>toggle fn<br>signature | redo<br>paste<br>register             | indent<br>line                | scroll line<br>down | half-page<br>down<br>delete to<br>bol | [Tab]        | previous<br>jump pos<br>ac copilot<br>word | next<br>jump pos<br>ac copilot<br>line | [Esc] |   | open<br>floating<br>terminal |    |
| Esc                   | A                            | S                                           | D                                     | F                             | G                   | H                                     | J            | K                                          | L                                      | ;     | ' | Enter                        |    |
|                       | aerial<br>view<br>del to eol | write buf<br>del prev<br>char               | half-page<br>up<br>unindent<br>line   | page up                       |                     | prev yank<br>entry                    | comment<br>▼ | comment<br>▲                               | next yank<br>entry                     |       |   | ac copilot all               |    |
| Z                     | X                            | C                                           | V                                     | B                             | N                   | M                                     | ,            | .                                          | /                                      |       |   |                              |    |
|                       | del next<br>char             |                                             | block<br>visual<br>paste<br>clipboard | page down<br>del prev<br>word | open<br>nvim-tree   | [Enter]                               |              |                                            | toggle<br>comment                      |       |   |                              |    |

in nvim-tree:  
? or g? - show mappings  
Enter - open file

**motion**

moves the cursor,  
or the screen

**edit**

changes the text

**open**

open a sidebar  
or floating window

**mode**

changes the nvim mode

|       |                                                     |
|-------|-----------------------------------------------------|
| Space |                                                     |
|       | open auto-complete menu<br>/ confirm autocompletion |

|  |             |
|--|-------------|
|  | normal mode |
|  | insert mode |

**visual mode:**  
Ctrl-C/X: copy/cut selection to register + (system clipboard)

**additional copilot commands:**  
Ctrl-Right: request suggestion  
Ctrl-Up/Down: next/previous suggestion  
Ctrl-Left: dismiss suggestion

**autocompletion:**  
Ctrl-Space / Tab: open autocompletion

in menu:  
Ctrl-j / Ctrl-k / Tab / Shift-Tab: move down/up  
Enter / Ctrl-Space: confirm suggestion  
Ctrl-U / Ctrl-D: scroll docs up/down  
Esc - close menu

in function snippet:  
Tab / Shift-Tab: next / previous placeholder

**in nvim-tree:**  
? or g? - show mappings  
Enter - open file  
v / Ctrl-v - open file in vertical split  
t / Ctrl-t - open file in new tab  
r - rename show full path  
Ctrl-R - rename omit base name  
Backspace - Close directory  
Tab - open preview (stay in nvim-tree)  
f / F - start/clear filter  
c / p - copy/paste node  
d - delete  
R - refresh view  
K - info about node  
Ctrl-] - change root  
a - create new file  
H / I - toggle hidden/gitignored files  
P - go to parent directory  
E / W or C - expand/collapse all

# NeoVim Cheat Sheet - [Alt]-Commands

|     |                         |                       |                 |                      |                    |                   |                         |   |                 |                       |   |                    |    |                    |   |                   |   |   |       |
|-----|-------------------------|-----------------------|-----------------|----------------------|--------------------|-------------------|-------------------------|---|-----------------|-----------------------|---|--------------------|----|--------------------|---|-------------------|---|---|-------|
| `   | 1                       | 2                     | 3               | 4                    | 5                  | 6                 | 7                       | 8 | 9               | 0                     | - | =                  | Bs |                    |   |                   |   |   |       |
| Q   | close all other windows | W                     | maximize window | E                    | rotate window left | R                 | rotate windows right    | T | move to new tab | Y                     | U | I                  | O  | P                  | [ | ]                 | \ |   |       |
| Esc | A                       | go to previous window | S               | equalize all windows | D                  | go to next window | F                       | G | H               | go to previous window | J | go to window below | K  | go to window above | L | go to next window | ; | ' | Enter |
| Z   | close window            | X                     | open new buffer | C                    | close tab          | V                 | split window vertically | B | N               | go to previous tab    | M | go to next tab     | ,  | .                  | / |                   |   |   |       |